

Subject Description Form

Subject Code	ITC3040T
Subject Title	Computer Technology for Fashion and Textiles
Credit Value	3 credits
Level	3
Pre-requisite/ <Co-requisite> / (Exclusion)	Nil
Objectives	The subject aims to provide students with an overview of computer technologies required for fashion and textile industry. Students will gain the ability of using computer aided design (CAD) and computer aided manufacture (CAM) in fashion and textile product development. By studying the general concepts of information system and working on various projects, students will possess the skill of integrating different computer systems and developing sound e-business strategies for fashion and textile industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Make use of the general computer technologies in fashion and textile industry; - Apply appropriate CAD systems in textile design or clothing pattern design; - Use and control CAM systems in fashion and textile manufacture. - Examine critically the use of computer systems that would meet the needs of a fashion business; - Communicate effectively with others regarding textile design, fashion design and pattern design and manufacturing systems.
Subject Synopsis/ Indicative Syllabus	(I) Overview of computer technology for fashion and textile CAD and CAM definition Functions of Textile and Clothing CAD Introduction to information system Integration of various information systems for e-business (II) Introduction of Textile CAD Overview of textile product development process Importance of computer aided design in fashion supply chain Introduction to graphic file and colour models (III) Application of Textile CAD Software Yarn design and simulation Woven, Knitted, and Printed fabric design Simulation of yarn, fabric colour and structure effects (IV) Introduction to Clothing CAD

	Software and hardware of apparel CAD/CAM Process flow of fashion product development Using computer systems in fashion product design and manufacturing (V) Application of Apparel CAD Software Fashion Design System Pattern input, pattern drafting, pattern modification and pattern output. Pattern grading Marker planning and material utilization 3D clothing simulation on virtual try-on, fitting evaluation and pattern alteration.																																													
Teaching/Learning Methodology	Lectures are designed to provide general outlines of key concepts of the subject and to provide guidance on further readings and applications. Videos may be used occasionally to facilitate teaching. Tutorials are designed to provide the hand-on practice for students' learning of the subject materials in order to understand the applications on different CAD and CAM systems in fashion & textiles design and manufacturing.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th></th></tr><tr><td>1. Assignment</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Project</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Upon completion of the course, students are expected to understand and master the skills and techniques for textile and fashion CAD/CAM. Such techniques must be acquired through extensive hand-on practices and exercises, therefore, the course is assessed by assignments. Students are also expected to be able to communicate effectively and think critically by applying CAD/CAM related techniques in their future career, such knowledge integration should be trained up through project work. In sum, assignments and projects are appropriate methods to assess students in achieving the intended learning outcomes.								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e		1. Assignment	60%	✓	✓	✓	✓	✓		2. Project	40%	✓	✓	✓	✓	✓		Total	100%						
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1. Assignment	60%	✓	✓	✓	✓	✓																																								
2. Project	40%	✓	✓	✓	✓	✓																																								
Total	100%																																													
Student Study Effort Expected	Class contact:																																													
	▪ Lecture		12 Hrs.																																											
	▪ Studio		26 Hrs.																																											
	Other student study effort:																																													
	▪ Assignments		67 Hrs.																																											
	Total student study effort		105 Hrs.																																											

**Reading List and
References**

Books

Zeid, I. (1991). CAD/CAM Theory and Practice. McGraw-Hill.
Gray, S. (1998). CAD/CAM in Clothing and Textiles. Aldershot: Gower Publishing.
Volino, P., Magnenat-Thalmann, N. (2000). Virtual Clothing. Berlin: Springer.
Colussy, M.K. (2001). Fashion Design on Computers. NY: Prentice-Hall.
Aldrich, W. (1992). CAD in Clothing and Textiles. London: BSP Publication.
Prima Vision User Manual.